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Reg. No. | | | | | | | | | | | [18ESCIC31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

DISCRETE MATHEMATICS AND GRAPH THEORY

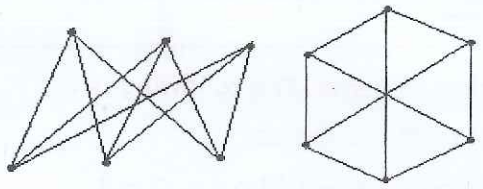
Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions
- **ii.** Questions 1 and 2 are compulsory.
- **iii.** Missing data may be suitably assumed.

1. a. Define the following terms with an example. **04**

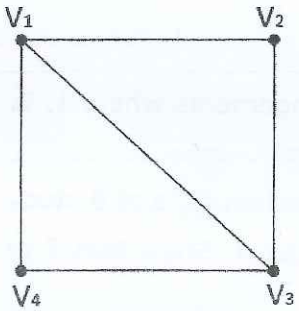
- **i.** Bipartite graph
- **ii.** Hamiltonian cycle

b. Define Graph Isomorphism. Verify whether the following graphs are Isomorphic or not. **05**



c. Define planar graph. A connected planar graph *G* has 9 vertices with degrees 2, 2, 3, 3, 3, 4, 5, 6, 6. Find the number of regions of *G*. **05**

2. a. Find the chromatic number and the chromatic polynomial for the graph given below. **04**



b. Find the number of permutations of the letters of the word MASSASAUGA. In how many of these, all four A's are together? How many of them begin with S? **05**

i. Find the coefficient of x^{12} in the expansion of $x^3(1-2x)^{10}$. **05**

ii. In how many ways can 10 identical pencils be distributed among 5 children in the following cases:

- **i.** There are no restrictions.
- **ii.** Each child gets at least one pencil.

3. a. Two marbles are drawn in succession from a box containing 10, 20 red , 20 blue and 30 white. Find the probability that first is red and second is white with no replacement after each drawing. **04**

b. A box contains 9 tickets numbered 1to 9 inclusive. If 3 tickets are drawn from the box one at a time, find the probability that they are alternatively either odd, even or even, odd, even. **05**

c. If A and B are two events with $P(A)=1/2$, $P(B)=1/3$ and $P(A\cap B)=1/4$, find $P(A|B)$, $P(B|A)$, $P(A|\overline{B})$, $P(\overline{A}|\overline{B})$ and $P(A\cup B)$. **05**

OR

4. a. Let $A=\{1,3,5\}$, $B=\{2,3\}$ and $C=\{4,6\}$. Determine **04**

i. $A\times B$

ii. $(A\cup B)\times C$

iii. $(A\times B)\cap(B\times C)$

iv. $(A\times B)\cap(B\times A)$

b. A professor has two dozen introductory text books on Computer Science and is concerned about their coverage of the topics **05**

(A) Compilers (B) Data Structures and (C) Operating systems. The following is the data on the number of books that contain material on these topics.

$|A| = 8,\ |B| = 13, |C| = 13, |A \cap B| = 5, |A \cap C| = 3,\ |B \cap C| = 6$ *and* $|A \cap B \cap C| = 2.$

i. How many have no material on Compilers.

ii. How many do not deal with any of the topics.

c. Let *f* and *g* be functions from *R* to *R* defined by $f(x)=ax+b$ and $g(x)=1-x+x^2$. If $(g\circ f)(x)=9x^2-9x+3$, determine *a*, *b*. **05**

5. a. For the positive integers 1, 2, 3, ...,*n*, there are 11660 derangements where 1, 2, 3, 4, 5 appear in the first five position . What is the value of *n*? **04**

b. Out of 30 students in a hostel, 15 study History, 8 study Economics, and 6 study Geography. It is known that 3 students study all these subjects. Show that 7 or more students study none of these subjects. **05**

c. By using the expansion formula, obtain the rook polynomial for the board C shown below(made up of unshaded parts): **05**

1		2
3	4	5
	6	

OR

6. a. There are eight letters to eight different people to be placed in eight different addressed envelopes. Find the number of ways of doing this so that at least one letter gets to the right person. **04**

b. In how many ways 5 number of *a*'s, 4 number of *b*'s and 3 number of *c*'s can be arranged so that all the identical letters are not in a single block? **05**

c. Four persons P_1, P_2, P_3, P_4 who arrive late for a dinner party find that only one chair at each of five tables T_1, T_2, T_3, T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2, P_2 will not sit at T_2, P_3 will not sit at T_3 or T_4, and P_4 will not sit at T_4 or T_5. Find the number of ways they can occupy the vacant chairs. **05**

7. **a.** Find the sequences generated by the function $3x^3 + e^{2x}$. **04**

b. Find the generating function of the sequences **05**

i. $1^2, 2^2, 3^2, 4^2, \ldots$

ii. $0^2, 1^2, 2^2, 3^2, \ldots$

c. Find the number of partitions of $n = 6$ into distinct summands. **05**

OR

8. **a.** Solve the recurrence relation **07**

$a_n - 3a_{n-1} = 5 \times 7^n$, for $n \geq 1$, given that $a_0 = 2$.

b. Solve the recurrence relation **07**

$a_n = a_{n-1} + a_{n-2}$, for $n \geq 3$, given that $a_0 = 2$, $a_1 = 3$.